

Lecture 8: The Product Rule

GOAL: Be able to differentiate functions of the form $f(x)g(x)$.

Summary:

X SinX

$$\frac{d}{dx}[f(x)g(x)] = \frac{df}{dx}g(x) + f(x)\frac{dg}{dx}$$

$$[f(x)g(x)]' = f'(x)g(x) + f(x)g'(x)$$

Think about why this is true or find f and g that makes it false

Question: Does $[f(x)g(x)]' = f'(x)g'(x)$? $\leftarrow$

$$f(x)=1$$

$$g(x)=x$$

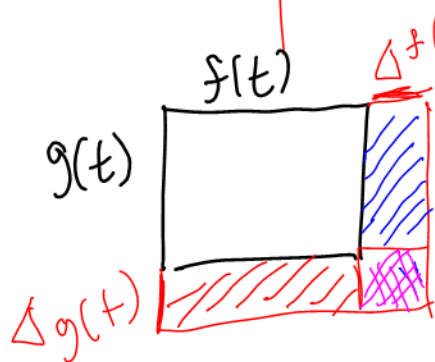
$$[f(x)g(x)]' = [x]'$$

$$= 1$$

$$f'(x) \cdot g'(x) = [1]' \cdot [x]'$$

$$= 0 \cdot 1$$

$$= 0$$



$$\Delta(f(t)g(t)) = \text{Area of Shaded Region}$$

$$= f(t)\Delta g(t) + \Delta f(t)g(t) + \left[\begin{matrix} \rightarrow 0 \\ \text{as} \\ \Delta t \rightarrow 0 \end{matrix} \right]$$

$$\frac{\Delta(f(t)g(t))}{\Delta t} = f(t)\frac{\Delta g(t)}{\Delta t} + \frac{\Delta f(t)}{\Delta t}g(t) + [\text{error}]$$

Theorem (Product Rule) Let f and g be diff. fns. Then

$$\frac{d}{dx}[f(x)g(x)] = \frac{df}{dx}g(x) + f(x)\frac{dg}{dx}$$

$$[f(x)g(x)]' = \underbrace{f'(x)g(x)}_{\text{right d-left}} + \underbrace{f(x)g'(x)}_{\text{left d-right}}$$

"right d-left plus left d-right"

Example 1 $\frac{d}{dx}(x^3) = 3x^2$

$$[x^3]' = [\underbrace{x}_f \cdot \underbrace{x^2}_g]' = [x]' \cdot x^2 + x \cdot [x^2]'$$

$$= 1 \cdot x^2 + x \cdot 2x = x^2 + 2x^2 = 3x^2$$

NOTE: By continuing Ex. 1, we can show $\frac{d}{dx}(x^n) = n \cdot x^{n-1}$
where n is an integer ≥ 0

$$x^4 = x \cdot x^3$$

$$x^5 = x \cdot x^4$$

...

Example 2 Find y' if $y = 2 \sin x \cos x$

$$\begin{aligned} y' &= [2 \sin x \cdot \cos x]' = [2 \sin x]' \cos x + 2 \sin x [\cos x]' \\ &= [2 \cos x] \cos x + 2 \sin x [-\sin x] \\ &= 2 [\cos^2 x - \sin^2 x] \end{aligned}$$

NOTE: $[2 \cdot \sin x \cos x]' = [2]' \sin x \cos x + 2 [\sin x \cos x]'$

Example 3 Let $h(x) = (x^{10} + x + 1) \cdot (x^2 + 9)$

@ Compute $h'(x)$

$$f(x) = x^{10} + x + 1 \quad \longrightarrow \quad f'(x) = 10x^9 + 1$$

$$g(x) = x^2 + 9 \quad \longrightarrow \quad g'(x) = 2x$$

$$\begin{aligned} h'(x) &= f'(x) g(x) + f(x) g'(x) \\ &= (10x^9 + 1)(x^2 + 9) + (x^{10} + x + 1)(2x) \end{aligned}$$

⑥ What's $h'(1)$?

$$h'(1) = 11 \cdot 10 + 3 \cdot 2 = 116$$

Example 4 Find h' if $h(x) = \underbrace{(2x+1)}_f \cdot \underbrace{(3x+1)(4x+1)}_g$,

$$\begin{aligned} h'(x) &= [2x+1]'(3x+1)(4x+1) + (2x+1)[(3x+1)(4x+1)]' \\ &= 2(3x+1)(4x+1) + (2x+1)[\underbrace{(3x+1)}_f \cdot \underbrace{(4x+1)}_g]' \\ &= 2(3x+1)(4x+1) + (2x+1)[3(4x+1) + (3x+1) \cdot 4] \\ &= 2 \cdot (3x+1)(4x+1) + (2x+1) \cdot 3 \cdot (4x+1) + (2x+1)(3x+1) \cdot 4 \end{aligned}$$

Example 5 Find the equation of the line tangent to $h(x) = \underbrace{(\sqrt{x}+4)}_f \cdot \underbrace{(x^{\frac{3}{4}} - 2x)}_g$ at $x=16$

Need to find $h'(16) = f'(16)g(16) + f(16)g'(16)$

$$f(x) = \sqrt{x} + 4$$

$$f(16) = \sqrt{16} + 4 = 4 + 4 = 8$$

$$f'(x) = \frac{1}{2}x^{-\frac{1}{2}} = \frac{1}{2\sqrt{x}}$$

$$f'(16) = \frac{1}{2\sqrt{16}} = \frac{1}{2 \cdot 4} = \frac{1}{8}$$

$$g(x) = x^{\frac{3}{4}} - 2x$$

$$= (\sqrt[4]{x})^3 - 2x$$

$$g(16) = 8 - 2(16) = 8 - 32$$

$$= -24$$

$$g'(x) = \frac{3}{4}x^{-\frac{1}{4}} - 2$$

$$= \frac{3}{4 \cdot \sqrt[4]{x}} - 2$$

$$g'(16) = \frac{3}{8} - 2 = -\frac{13}{8}$$

$$h'(16) = f'(16)g(16) + f(16)g'(16)$$

$$\uparrow = \frac{1}{8} \cdot (-24) + 8 \cdot \left(-\frac{13}{8}\right)$$

$$\text{Slope} = -3 - 13 = -16$$

Point : x -coord : 16

$$y\text{-coord: } h(16) = f(16)g(16) = 8(-24) = -192$$

Eq:

$$y - h(16) = h'(16)(x - 16)$$

$$y + 192 = -16(x - 16)$$

$$y = -16x + 64$$

Ex 6 When does $y = \underbrace{10x^5}_f \underbrace{e^x}_g$ have a horizontal tangent line?

$$y' = 50x^4 e^x + 10x^5 \{e^x\}' \leftarrow \text{Technically } [e^x]'$$

$$50x^4 e^x + 10x^5 e^x \stackrel{\text{set}}{=} 0$$

$$10x^4 e^x (5 + x) = 0$$

$$\begin{array}{l} 10=0 \\ \text{Never} \end{array} \quad \swarrow \text{OR} \quad \boxed{x^4=0} \quad \text{OR} \quad e^x=0 \quad \text{OR} \quad \boxed{5+x=0}$$

Never

Example 7 Compute $h'(x)$ if

$$h(x) = 6e^x \sin x - 13e^x \cos x$$

$$= \underbrace{e^x}_f \underbrace{(6\sin x - 13\cos x)}_g$$

$$h'(x) = [e^x]' (6\sin x - 13\cos x) + e^x [6\sin x - 13\cos x]'$$

$$\begin{aligned}h'(x) &= e^x(6\sin x - 13\cos x) + e^x(6\cos x + 13\sin x) \\&= e^x(6\sin x - 13\cos x + 6\cos x + 13\sin x) \\&= e^x(19\sin x - 7\cos x)\end{aligned}$$

$$\left[\frac{f(x)}{g(x)}\right]' = \left[f(x) \cdot \frac{1}{g(x)}\right]'$$